

Eliminating the Aleatoric Ceiling in

Ground-Level NO₂ Spike Prediction

Progress report — methodology and experiments to date

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JPL SURF Intern · Summer 2024

Stations

69

Satellite

TEMPO

Model

XGBoost

Features

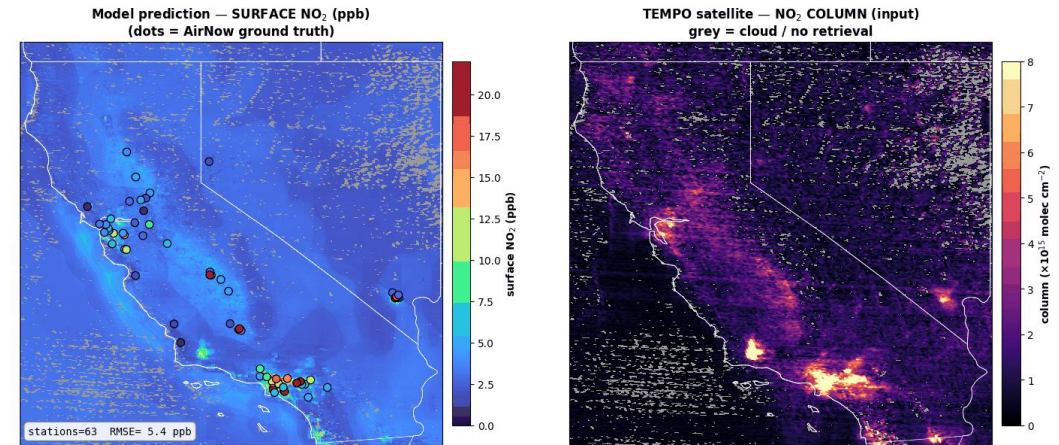
34

What Is TEMPO?

NASA's first geostationary air-quality satellite.

- Stares continuously at North America — not a polar orbit
- Scans every 1–2 hours during daylight (vs. once/day for older satellites)
- Measures column NO_2 over California & Nevada, Aug 2023 – Sep 2024

2024-07-10 13:17 PDT (20:17 UTC)



Why this matters for the project:

TEMPO sees the whole sky, but ground monitors see what people breathe. Our job is to learn the bridge between the two, so we can eventually map surface pollution anywhere TEMPO looks, not just at the 69 monitor sites.

Why Do NO₂ Spikes Matter?

A 'spike' = NO₂ ≥ 30 ppb at any of the 69 ground monitors.

2.5%

of all readings
are spikes

52%

of spikes come
from just 3 stations

107 ppb

highest reading
ever observed

Spikes cluster near industrial corridors as expected, they are physical events tied to real sources. Predicting them matters because acute exposure near these hotspots is a real public-health concern, as millions of people live there.

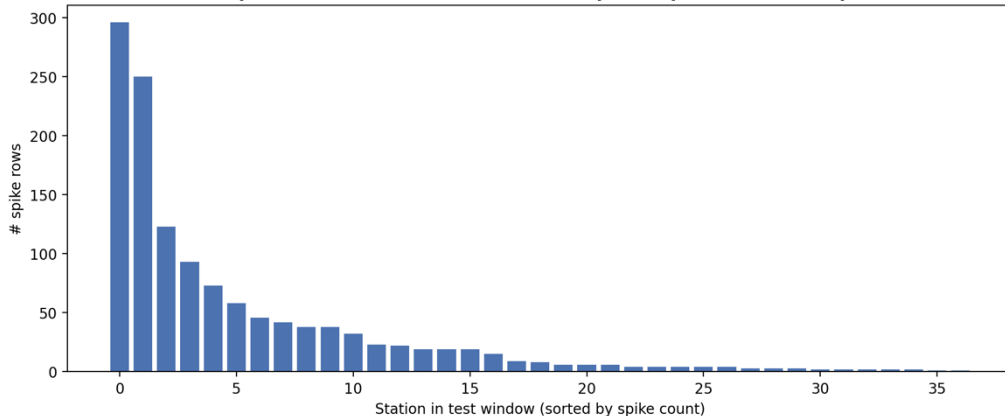
TOP STATIONS:

60 Near Road — 296 spikes (23.1%)

Ontario Near Road — 250 spikes (19.5%)

Fontana - Arrow Highway — 123 spikes (9.6%)

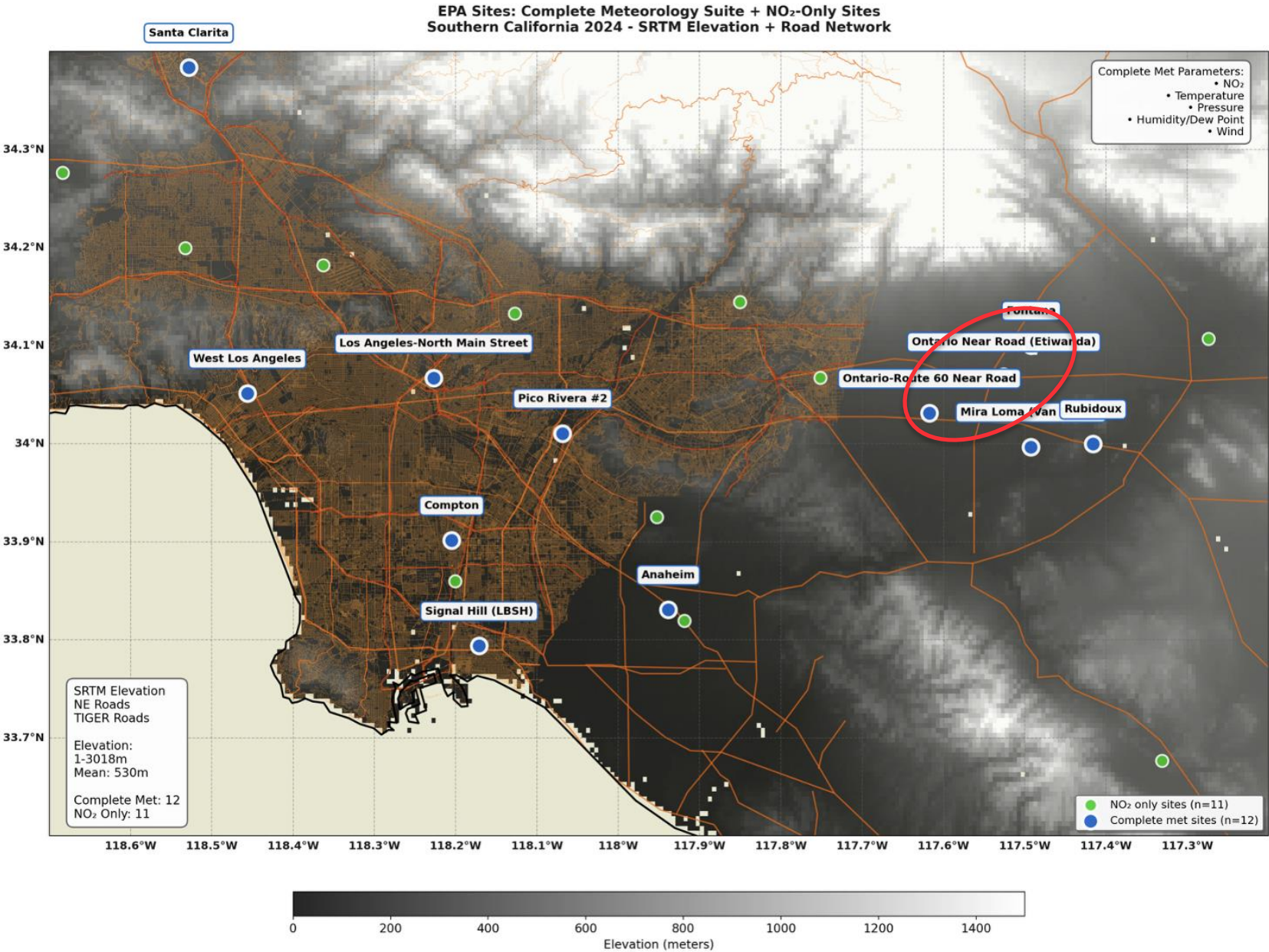
Are spikes concentrated? 37/69 stations spike; top 3 = 52% of all spikes



Why 30 ppb?

Not an EPA/NAAQS health limit (that's 100 ppb) — it's an empirical cutoff from our own data. The training mean is 7.8 ppb and median is 4.8 ppb; 30 ppb sits at the 97.5th percentile, i.e. roughly 4× the average reading. It marks a natural 'long tail' line: rare enough to represent a true outlier event, common enough (~4,500 rows) to still model and evaluate.

LA Basin: Complete Meteorology + NO₂-Only Sites with SRTM Elevation



Three Data Sources, One Training Table

TEMPO

Satellite column NO₂

1 feature

ERA5

Weather reanalysis
(wind, temp, pressure...)

7 features

Static + Time

Land use, roads, population,
hour/month/day-of-week

26 features

1 + 7 + 26 = 34 features total, joined by station location and timestamp

All 69 EPA monitoring stations across California and Nevada are represented in both the training and test sets — we split by TIME, not by location (next slide).

The 7 Meteorology Features (ERA5)

Weather drives whether pollution stays trapped near the surface or disperses. These 7 variables, resampled to each TEMPO scan time and station location:

t2m

2 m air temperature

d2m

2 m dew point

u10 / v10

10 m wind components (speed + direction)

sp

Surface pressure

blh

Boundary layer height — how 'trapped' pollution is

tcc

Total cloud cover

TAKEAWAY

blh (boundary layer height) is the most physically important of these — a shallow layer traps NO₂ near the ground, and it just gets more concentrated.

The 26 Spatial & Temporal Features

Location & Land Use (18)

Population density · Elevation · Traffic · Road density
Distance to coast · Distance to major roads
WorldCover land class (built / tree / vegetation / crop / barren / water)
Land-use category (urban / suburban / rural)
Station region (SoCal / Central / Upper CA)

Time & Sun Position (8)

Hour of day (UTC) (sin/cos) · Month of year (sin/cos)
Day of week (sin/cos) · is_weekend
Solar zenith angle

Why cyclic (sin/cos) encoding? Hour 23 and hour 0 are one hour apart in reality but far apart as raw numbers — sin/cos wraps them correctly.

Train / Test Split: By Time, Not by Location

TRAIN

July 2023 – Jun 2024 (11 months)
182,861 rows · 69 stations

How much missing data is there

TEST

Jul – Sep 2024 (3 months)
56,919 rows · same 69 stations

TAKEAWAY

All 69 stations appear in BOTH train and test — the model has seen every location before, but never these specific dates. This avoids data leakage and mirrors how the model would actually be deployed.

Spike rows: 4,496 in train (2.5%) · 1,282 in test (2.3%) — the imbalance makes this problem hard.

Why XGBoost?

Best-in-class for tabular data — rows of weather/satellite numbers are exactly its strength.

Captures non-linear physics a straight line can't — e.g. NO_2 falls off roughly as $1/\text{wind}$ (exponential scouring).

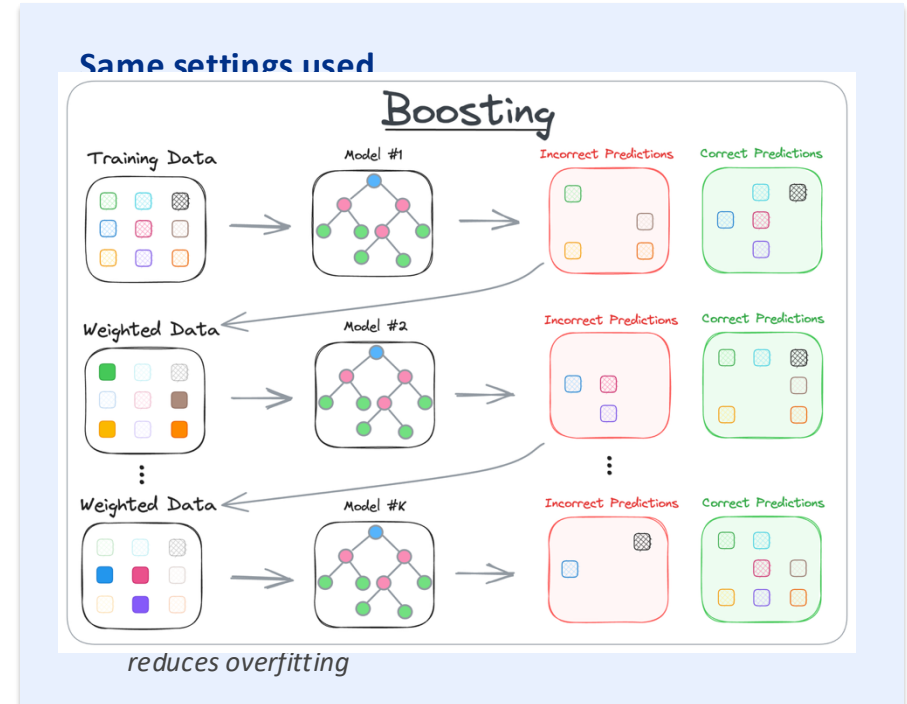
Beat a neural-network MLP on the same honest test:

– **XGBoost $R^2 = 0.35$ vs. Neural-net MLP $R^2 = 0.33$**

Beats classical baselines (persistence, seasonal climatology, wind-only linear regression).

Fast on CPU, handles missing values, robust to feature scale → simple **Yes/No** logic conditions on the feature values

Interpretable — we can show WHY it predicts what it predicts (feature importance + SHAP).



Keeping hyperparameters fixed means any change in results across experiments comes from the DATA, not from re-tuning the model.

How We Train It

14 months of data, tuned carefully

- Data: 14 months (July 2023 – September 2024) of hourly TEMPO + weather matched to EPA stations.
- Model uses 34 features
- We fixed XGBoost hyperparameters to 400 trees, depth 6, learning rate 0.05, and L2 regularization $\lambda=5.0$
 - These were derived from initial tuning on initial baseline models, applied across all 34-feature experiments
 - Maintains methodological rigor and avoid overfitting to retuning
 - Did NOT re-run RandomizedSearch for each feature engineering iteration (keeping hyperparameters fixed isolates the contribution of new features)
- **The current model uses 34 spatio-temporal features; 400 trees, depth 6, learning rate 0.05, regularization $\lambda=5$.**
- Reproducible: fixed random seed (42); model + metadata saved to disk.

Training data

Training rows

~182,900

Test rows

~56,900

Training stations

69

Time span

11 months train / 3 months test

Final features

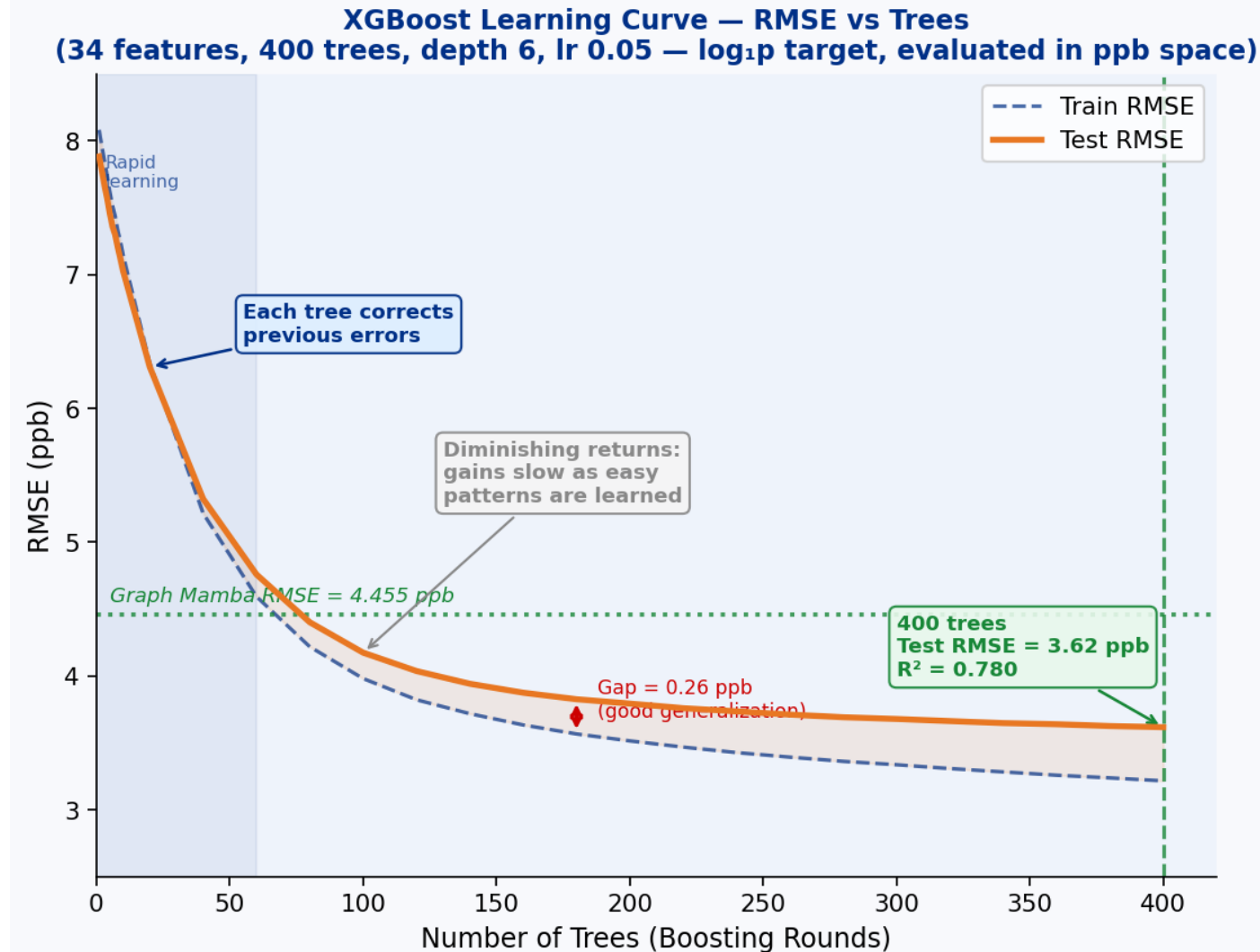
34 (spatio-temporal)

Random seed

42 (reproducible)

XGBoost's Real “Loss Curve”: RMSE vs. Trees

Error drops as trees are added



How XGBoost Works

(Gradient Boosted Trees)

① Start

Predict mean NO₂
for every row

② Residuals

Compute errors
(actual – predicted)

③ Tree 1

Fit a decision tree to the
residuals using 34 features

④ Trees 2-400

Each new tree corrects what
the last one missed
(lr = 0.05 scales contribution)

⑤ Final sum

Sum all 400 trees
→ RMSE 3.62 ppb $R^2 = 0.78$

The Starting Problem: Baseline Underprediction

The baseline model is accurate on average — and nearly blind to spikes.

+10.4 ppb

average spike
underprediction

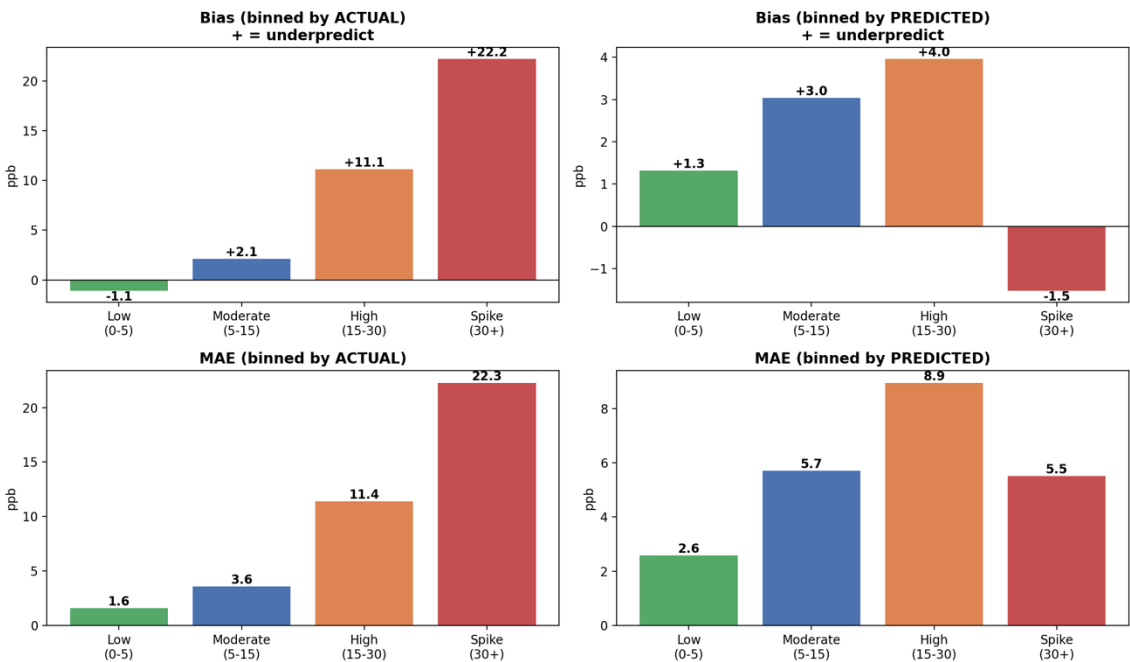
37.6%

of real spikes
ever caught

62 ppb

highest prediction ever made
(true max: 107 ppb)

Where the error lives: NO2 level bins, two honest views



Left: error grows sharply above ~20 ppb. This is because when 97.5% of the training data is 'normal,' the model learns to always guess close to normal.

Three Reasons the Model Underpredicts

1

The loss function itself

Standard training (MSE) mathematically aims for the AVERAGE outcome, not the extreme one. It is working as designed, just isn't designed for the goal of getting spikes.

2

97.5% of the data is 'boring'

Every training update is dominated by non-spike rows. The rare spike signal gets drowned out before the model can learn from it.

3

We're missing source data

We know region and land-use, but not which factory is running hard right now, or which way the wind is blowing from it.

TAKEAWAY

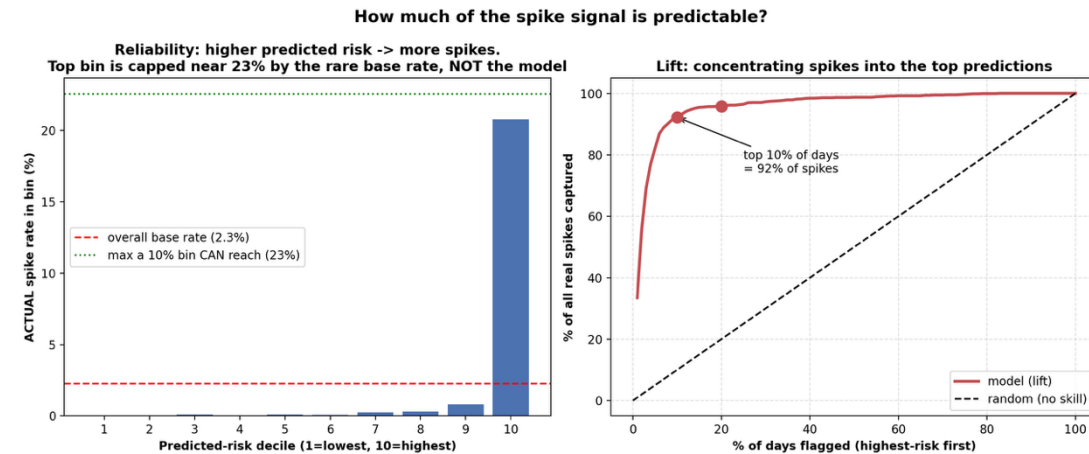
Each of these three causes gets its own experiment in the pages ahead: loss function → lagged features → real emissions data.

How Do We Know If We're Actually Improving?

Our metric: Top-1% Purity

1. Ask the model: "how likely is a spike today?" for every reading
2. Take its 1% most-confident predictions
3. Check: how many of those are REAL spikes?

Higher purity = if the model can predict extreme spikes, the model genuinely understands spike risk, not just guessing the average.



R^2 (the standard accuracy score) is misleading here — a model that always predicts 'normal' scores well on R^2 , while being useless for spikes.

BASELINE

Starting point (our baseline model): For the Top-1% Purity it hit 75%

Our Experiment Roadmap

Each experiment attacks one root cause. Every prediction was written down BEFORE running it.

1	Change the loss function	76%
2	Add yesterday's NO ₂ reading	78%
3	Add station climatology	78%
4	Rebalance training toward spikes	74–78%
5	Add real power-plant emissions	79%

Top-1% Purity (started at 75%)

Experiment 1: Change the Loss Function

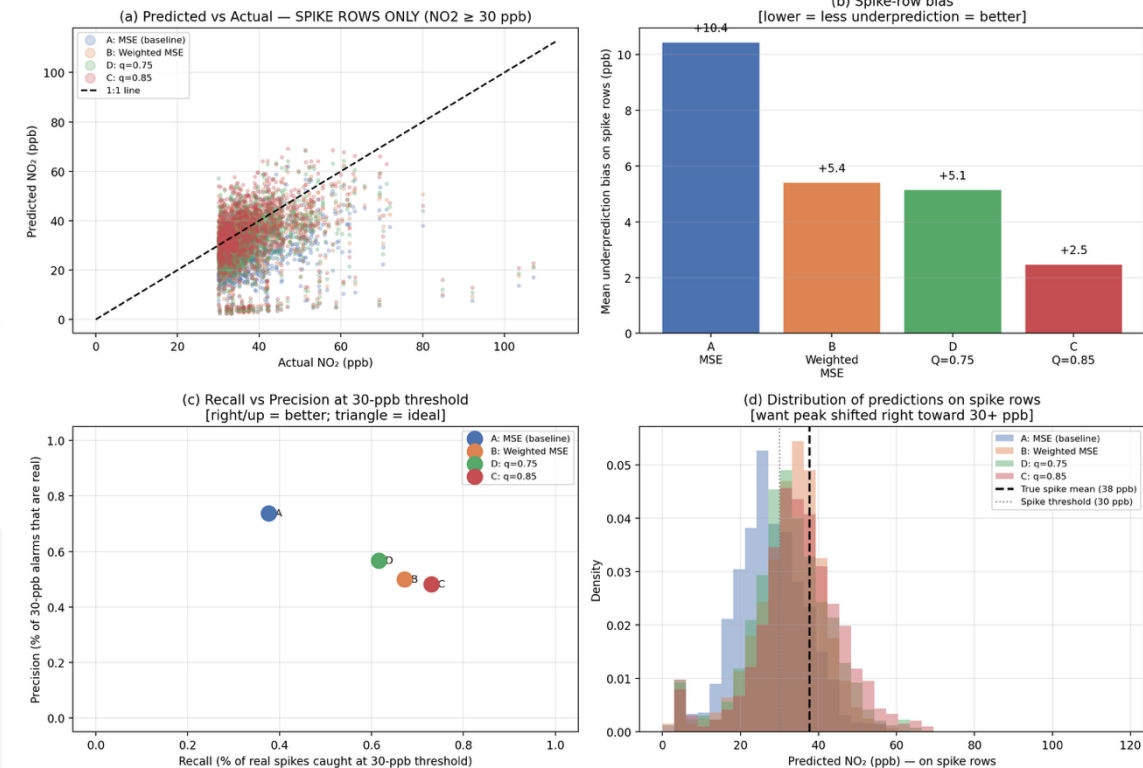
Idea: stop training the model to predict the AVERAGE.

- Switched to 'quantile regression' — model targets the 85th percentile, not the mean
- Also log-transformed NO_2 so huge spikes don't get ignored during training
- Our baseline Top-1% Purity started at 75% (and bumped to 76% here), The standard MSE of our baseline loss was underpredicting actual spikes by an average of +10.4 ppb, catching only 38% of real spikes.

TAKEAWAY

By switching to quantile regression, spike underprediction fell 76% (+10.4 → +2.5 ppb). We now catch 73% of spikes (up from 38%). This became our new baseline going forward.

Loss-Function Experiment: Does changing the objective reduce underprediction?
(Temporal split — known stations, test set only)



Experiment 2: Give the Model a Memory

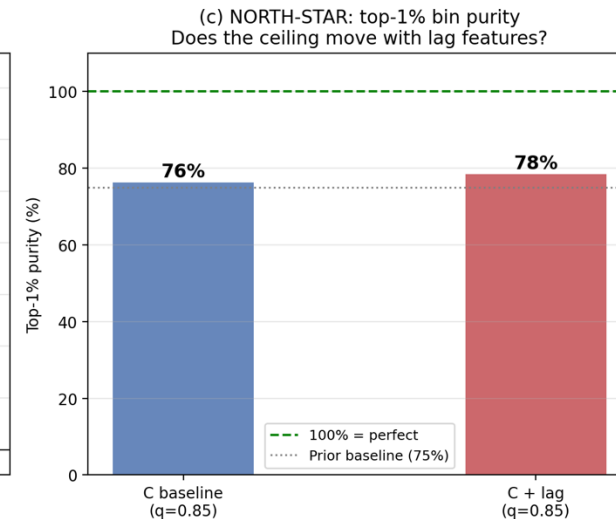
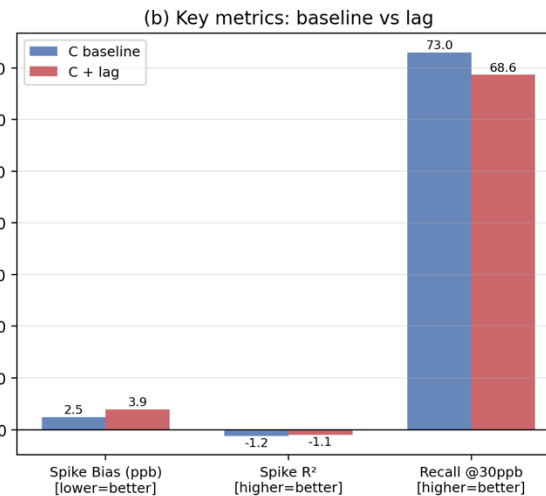
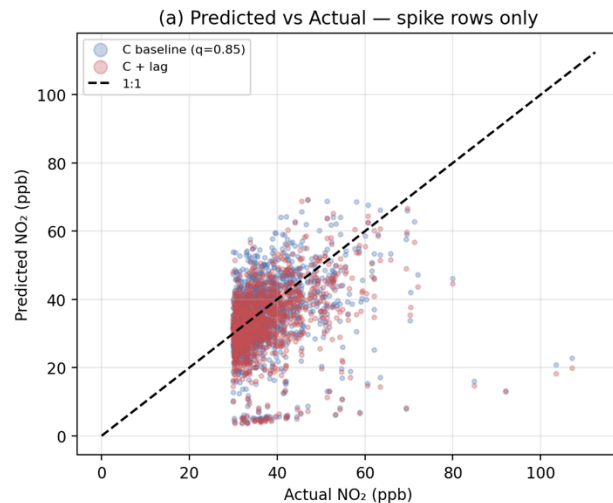
Idea: pollution episodes last for days, not hours.

- Added "yesterday's NO₂ reading" at each station as a new feature
- NO₂ lingers in the air 12–24 hours — high yesterday means high risk today

TAKEAWAY

This became our #1 most important feature overall. Top-1% purity rose 76% → 78%, meaning this information was missing before and is important!

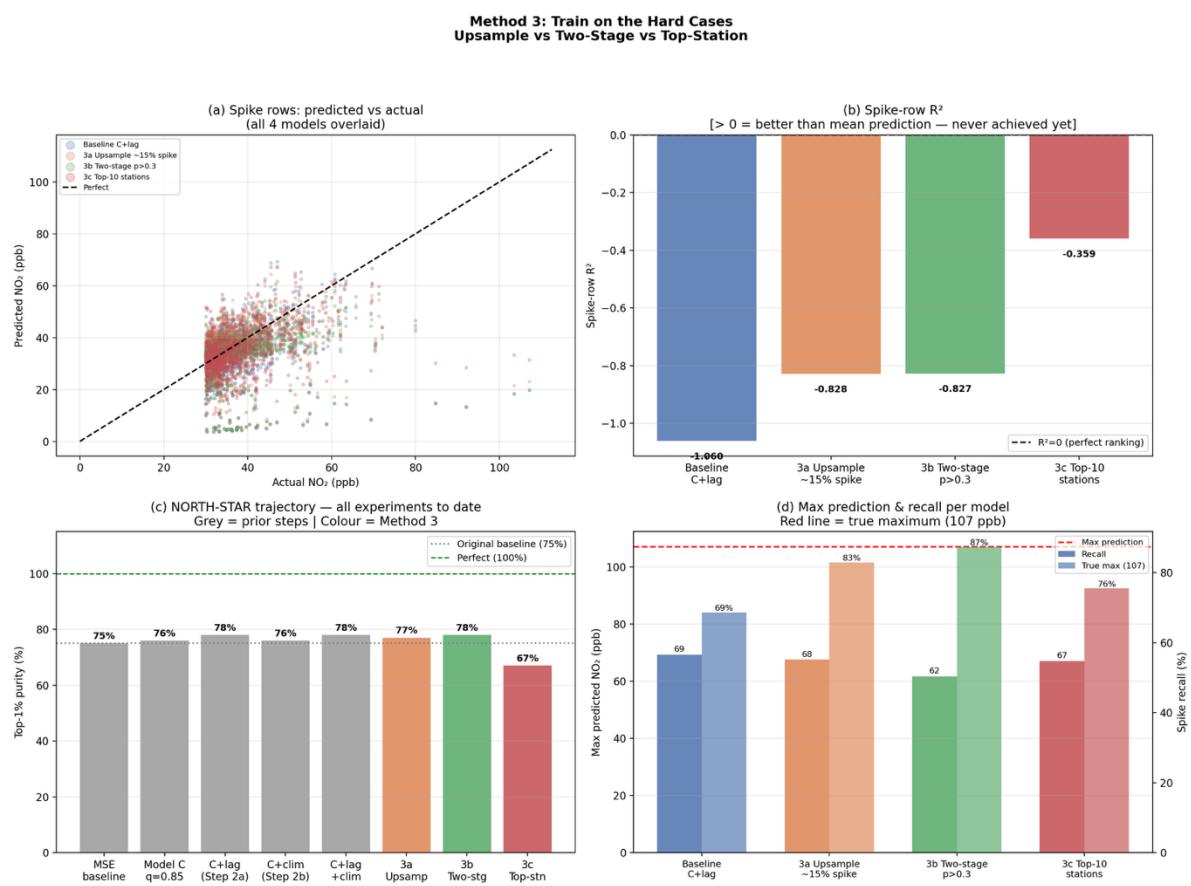
Step 2: Do lag features reduce spike underprediction?
(Temporal split — all 69 known stations)



Experiment 3: Rebalance Training Toward Spikes

Idea: give spikes more ‘weight’ during training so they aren’t drowned out.

- Tried 3 approaches: duplicating spike rows, a two-stage spike-only model, and training only on the highest-spike stations
- Also added each station’s own historical spike rate as a feature



Most important finding:

Training ONLY on our 10 most spike-prone stations produced by far our best spike-magnitude score ever — proving the signal genuinely exists in the data.

But that same model failed badly everywhere else. It only works if we can tell in advance which stations are high-risk.

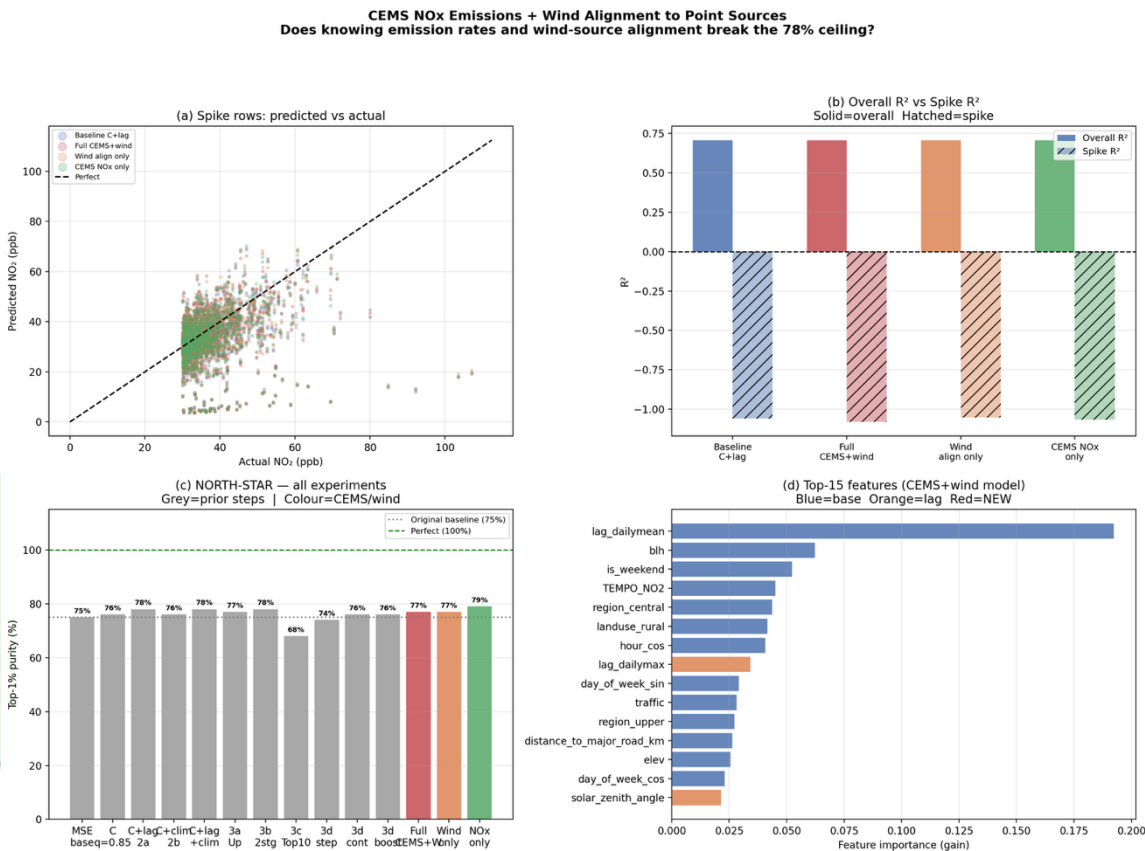
Experiment 4: Add Real Power-Plant Emissions

Idea: use actual EPA smokestack sensors near our stations.

- EPA requires large power plants/refineries to report hourly stack emissions
- Added: how much NOx nearby plants emitted, and whether wind blew from them toward the monitor

TAKEAWAY

Emissions data alone reached our best-ever purity score: 79%. Wind-direction alignment didn't help much — likely needs a tighter search radius around known plants, which is a clear next step.



Experiment 4 continued

Baseline C + lag: Top-1% **78%**, spike R^2 **-1.060**

5 km: Top-1% **78%**, spike R^2 **-1.080**

10 km: Top-1% **75%**, spike R^2 **-1.020** (*best magnitude fit / lowest MAE*)

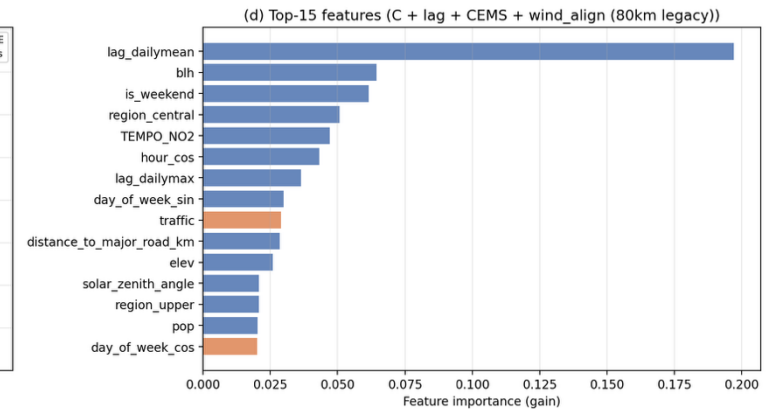
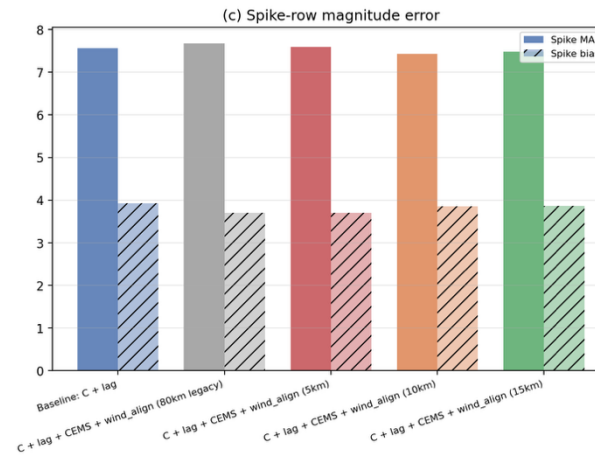
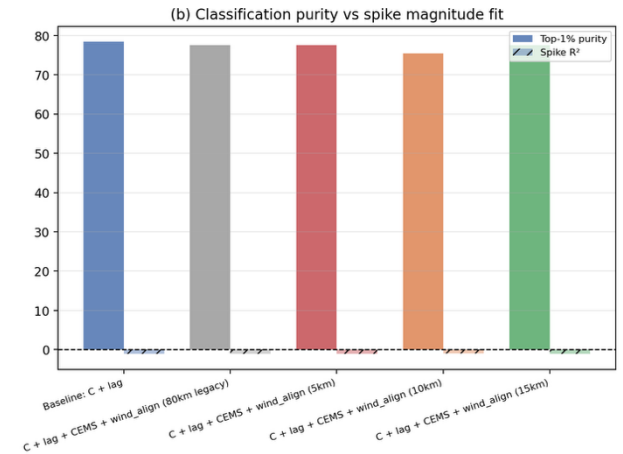
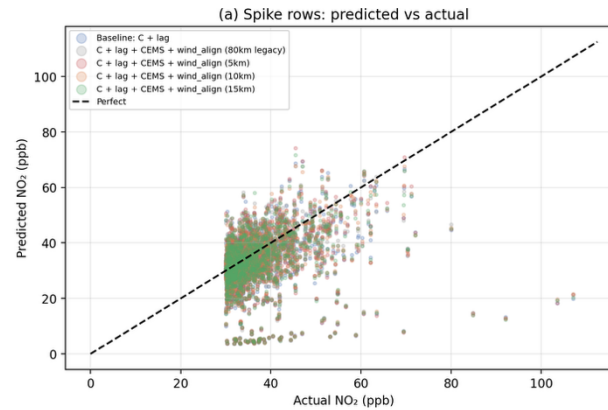
15 km: Top-1% **78%**, spike R^2 **-1.042**

Legacy 80 km: Top-1% **78%**, spike R^2 **-1.088**

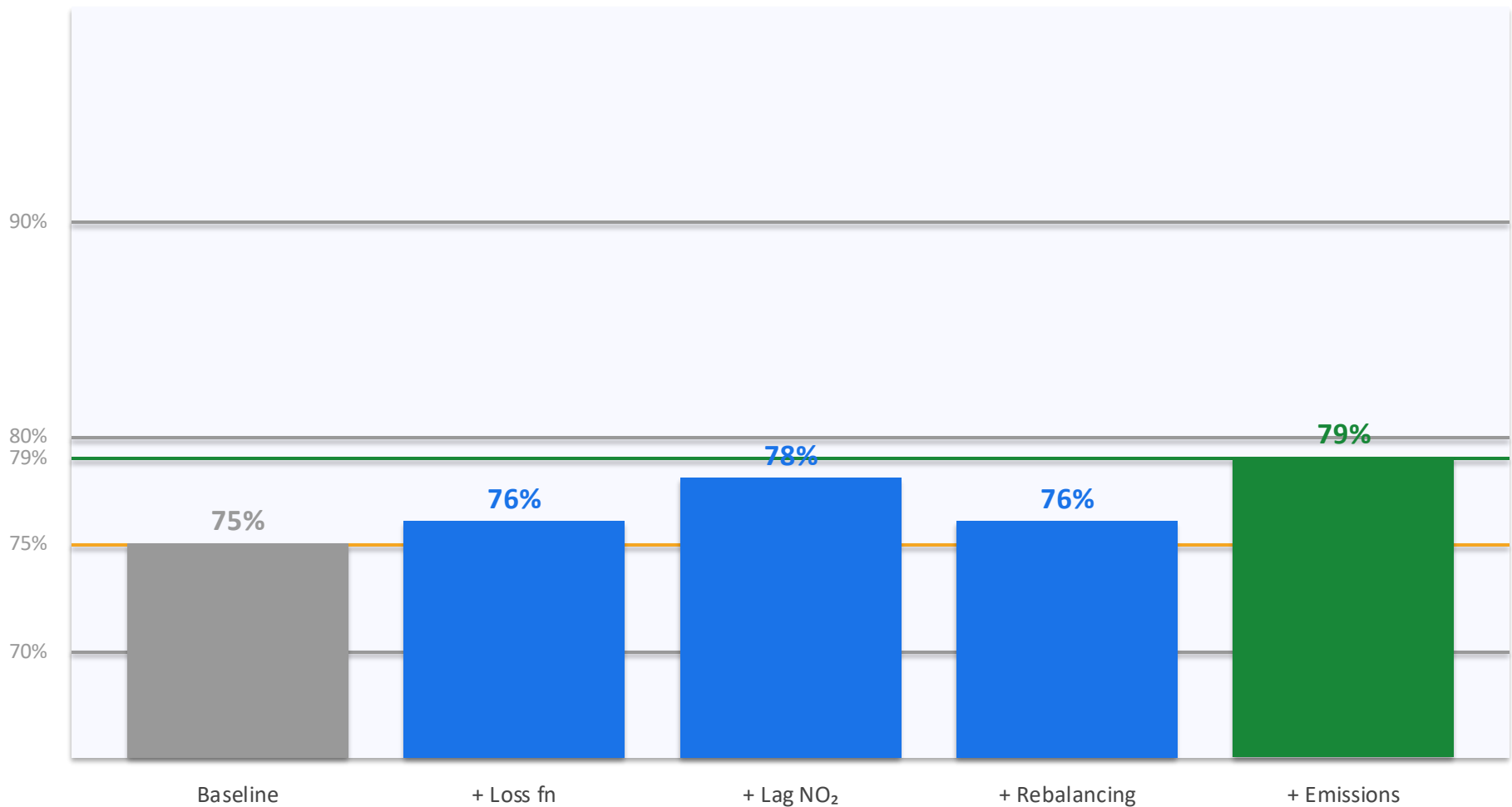
TAKEAWAY

So: tighter CEMS radii help spike magnitude a bit, but not Top-1% purity yet.

CEMS NO_x Emissions + Wind Alignment to Point Sources
Do tighter spatial buffers improve spike purity and spike magnitude?



Overall Progress: Top-1% Purity



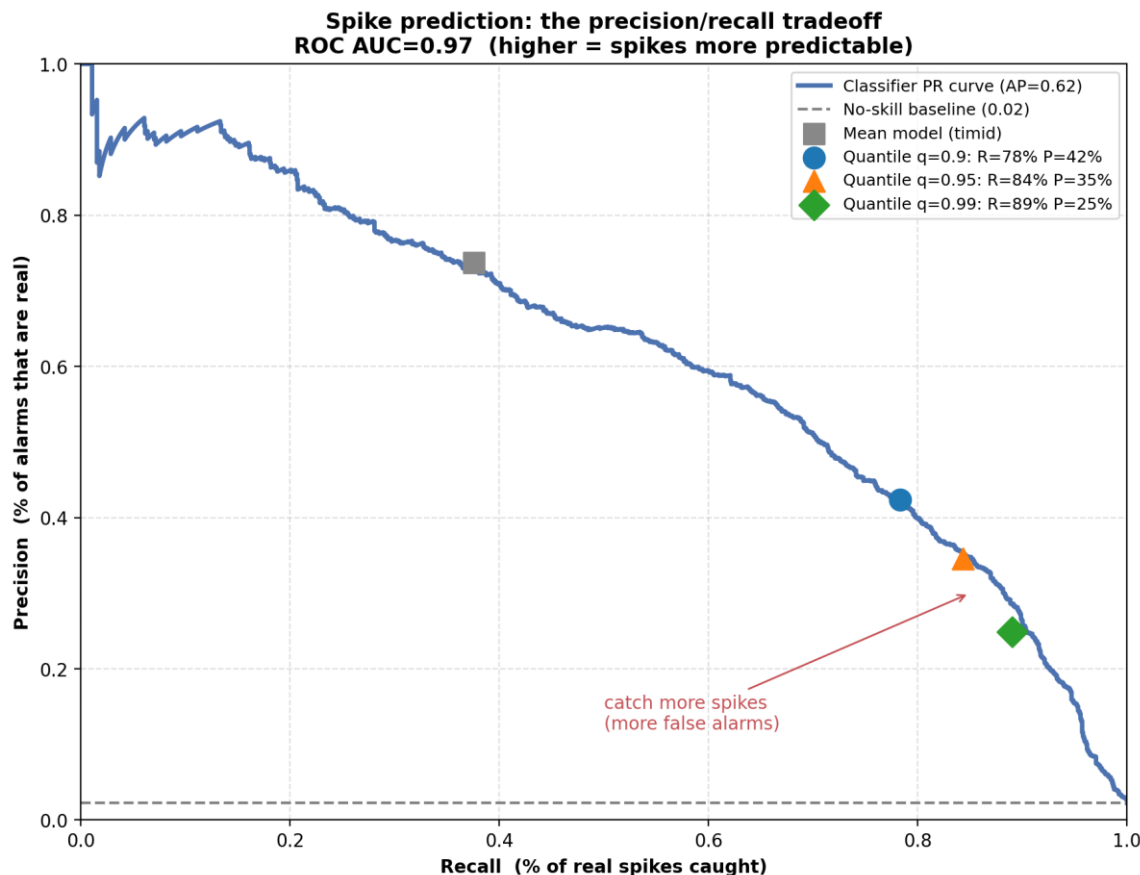
4 percentage points gained

75% → 79%

Each point represents real spikes the model now catches that it used to miss entirely.

Where We Still Fall Short

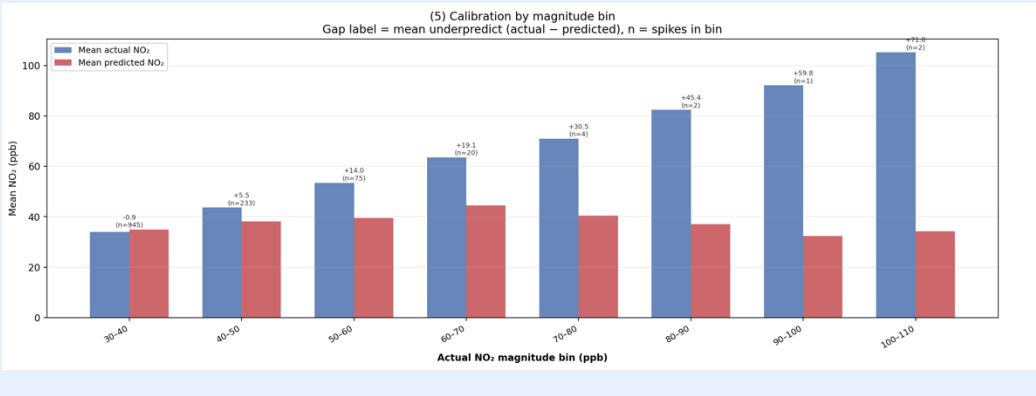
We can now tell WHEN a spike is likely (79% purity). We still struggle to predict HOW BIG it will be.



What this means:

Magnitude prediction only worked well in one narrow test (top-10 spiking stations only) which proves the signal exists, but we don't yet have features detailed enough to use it everywhere.

However, it is still a far away leap for us, as we can see we are still being diminished by the average:



Summary & Next Steps

What we've learned so far:

- The loss function mattered a lot
- Yesterday's reading is the single best predictor we've found
- Spike magnitude IS learnable to an extent, but mainly we are stuck to predict when a spike will happen

What we want to try next:

- Something
- Targeted wind-direction search around known plants
- Wildfire smoke flags for extreme outlier spikes

Current best model

79%

Top-1% Purity

70.4%

Spike Recall

+3.8 ppb

Spike Bias (was +10.4)

Thank you!

Comparison: Our XGBoost vs. Graph Mamba

How does our spike-detection pipeline stack up against the Graph Mamba interpolation approach?

Different tasks • different data ranges • complementary strengths

What Each Model Is Trying to Do

My Approach: XGBoost Spike Detection

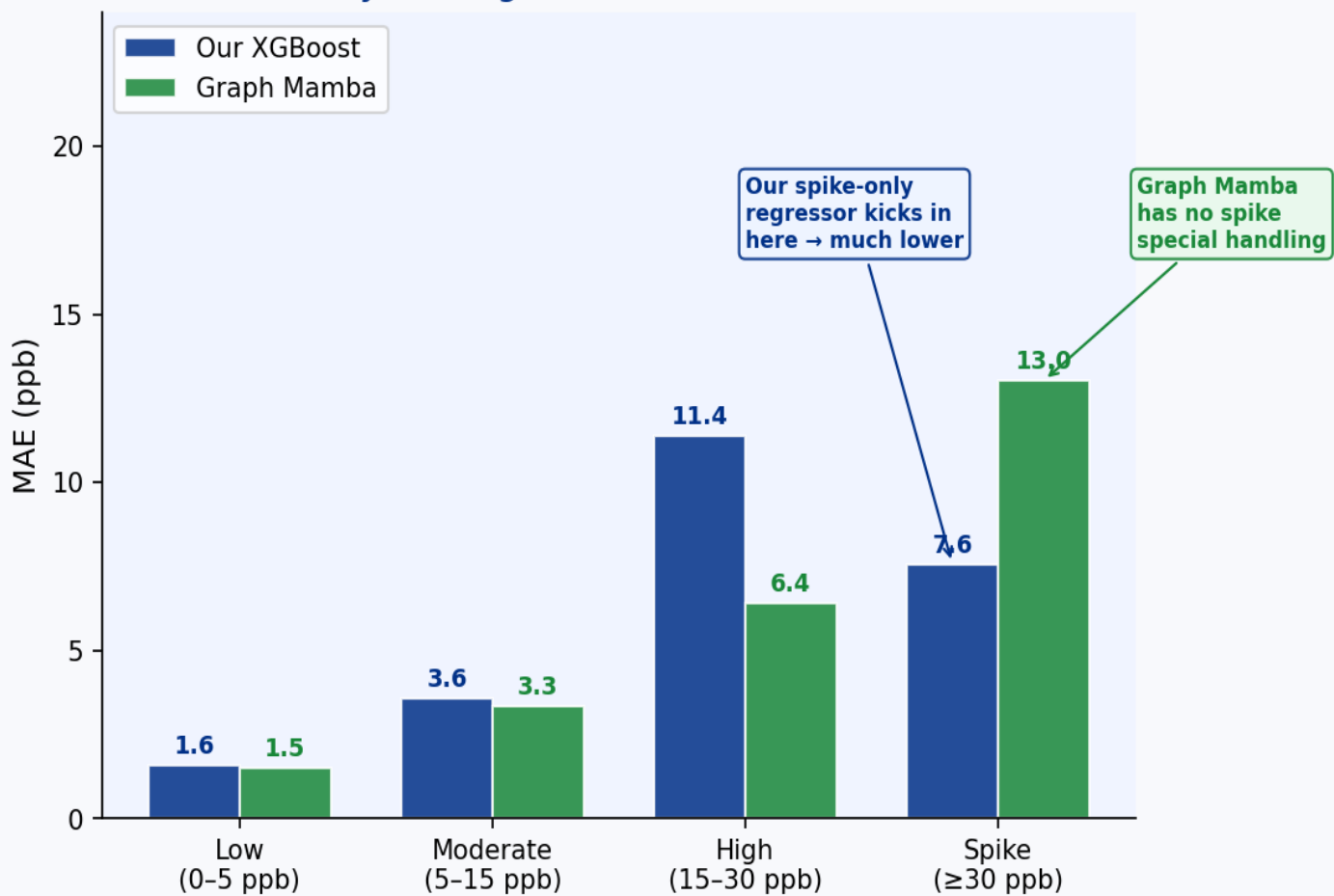
- **Task:**
Predict whether (and when) a NO₂ spike ≥ 30 ppb will occur
- **Data:**
July 2023–Sep 2024, 69 AirNow stations, California
- **Method:**
XGBoost classifier + spike-only regressor (hurdle model)
- **Key features:**
Lag NO₂, emissions (CEMS), station climatology, meteorology
- **Best result:**
79% Top-1% Purity | 70.4% Spike Recall
- **Magnitude bias:**
+3.8 ppb on spike rows (was +10.4 ppb at baseline)

Kyo's Approach: Graph Mamba Interpolation

- **Task:**
Spatially interpolate NO₂ from TEMPO satellite to AirNow sites
- **Data:**
July–Sep 2024, 191 stations, North America
- **Method:**
Graph Neural Network (Mamba) using TEMPO scan-aligned observations
- **Key features:**
Spatial graph structure, TEMPO pixel proximity, temporal gaps (δt)
- **Best result:**
RMSE 4.455 ppb | MAE 2.640 ppb | R² 0.594
- **Bias:**
+0.171 ppb overall (slight overprediction)

Head-to-Head: How Our Results Compare

MAE by NO₂ Regime: Where Each Model Wins



Head-to-Head Scorecard

Metric	Our XGBoost	Graph Mamba	Winner
Overall R ²	0.707	0.594	ours ✓
Overall RMSE (ppb)	4.18	4.455	ours ✓
Overall Bias (ppb)	~0	+0.17	tie ≈
Low NO ₂ MAE	1.58	1.52	GM ✓
Moderate NO ₂ MAE	3.58	3.34	GM ✓
High NO ₂ MAE	11.38	6.39	GM ✓
Spike MAE	7.56*	13.02	ours ✓
Spike bias	+3.93	-12.26	ours ✓
Max predicted	~70 ppb	51 ppb	ours ✓
Training time	< 5 min	~144 hrs	ours ✓
# Stations	69 (CA)	188 (N.Am.)	GM ✓
# Features used	34	6 + graph	different

* Spike MAE uses spike-only regressor (Stage 2), not full model

Takeaway: At low-moderate NO₂ Graph Mamba leads slightly; at spikes our two-stage model cuts MAE nearly in half (7.6 vs 13.0 ppb). Our model trains in minutes on a single GPU vs ~144 GPU-hours.

Graph Mamba: Overall Performance (Jul–Sep 2024)

4.455 ppb

RMSE

2.640 ppb

MAE

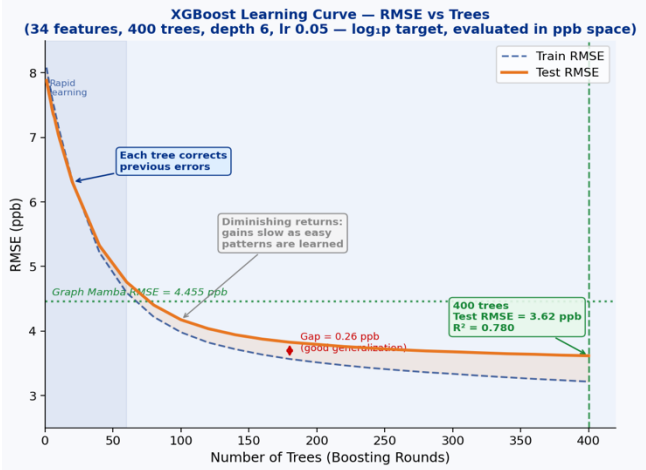
$R^2 = 0.594$

Variance explained

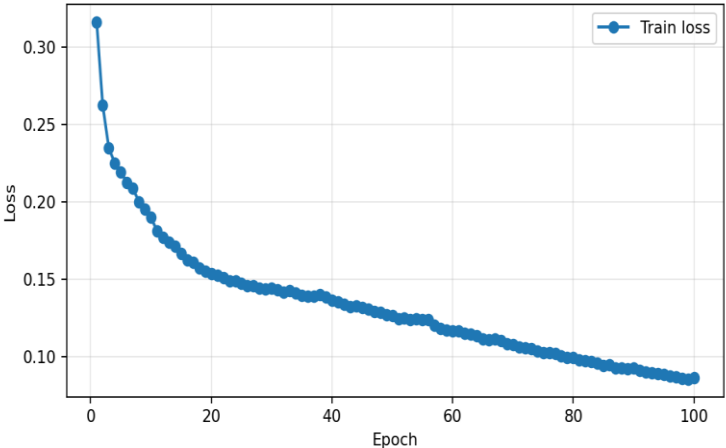
+0.171 ppb

Mean bias

XGBoost:



Training Loss by Epoch



GraphMamba:

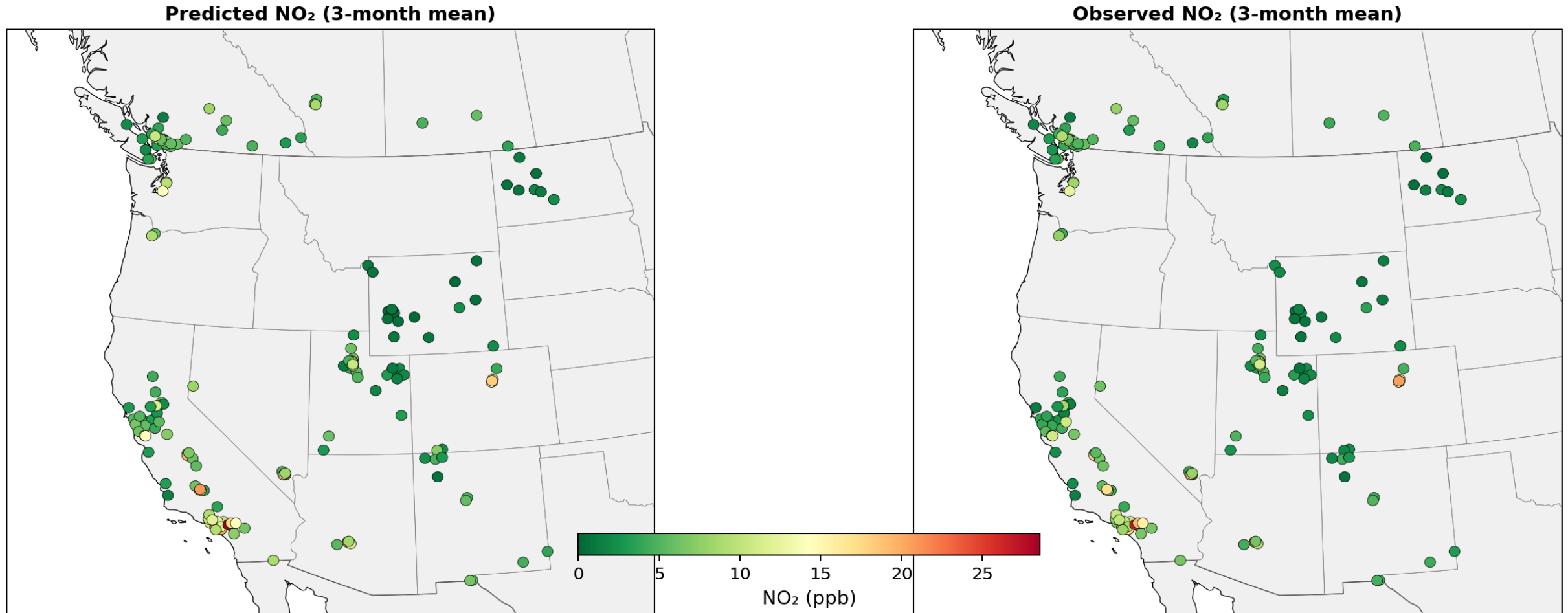
Error by Concentration Regime

Regime	n obs.	MAE	RMSE
< 5 ppb	107,615	1.52	2.69
5–15 ppb	54,166	3.34	4.57
15–30 ppb	16,138	6.39	7.92
≥ 30 ppb	2,887	13.02	16.37

High-concentration events (≥ 30 ppb) are hardest to predict — MAE jumps to 13 ppb in that regime, similar to our own spike-magnitude challenge.

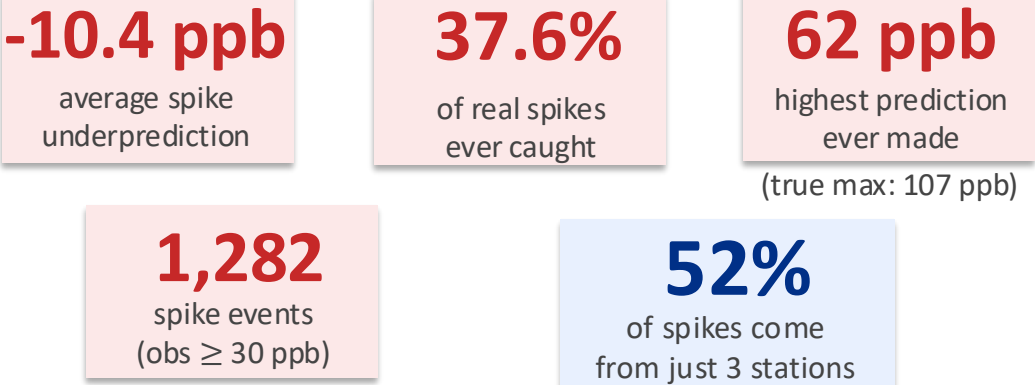
Graph Mamba: Spatial NO₂ Mean — Predicted vs. Observed

**Predicted vs Observed NO₂ at AirNow Stations
Jul - Sep 2024 Mean**

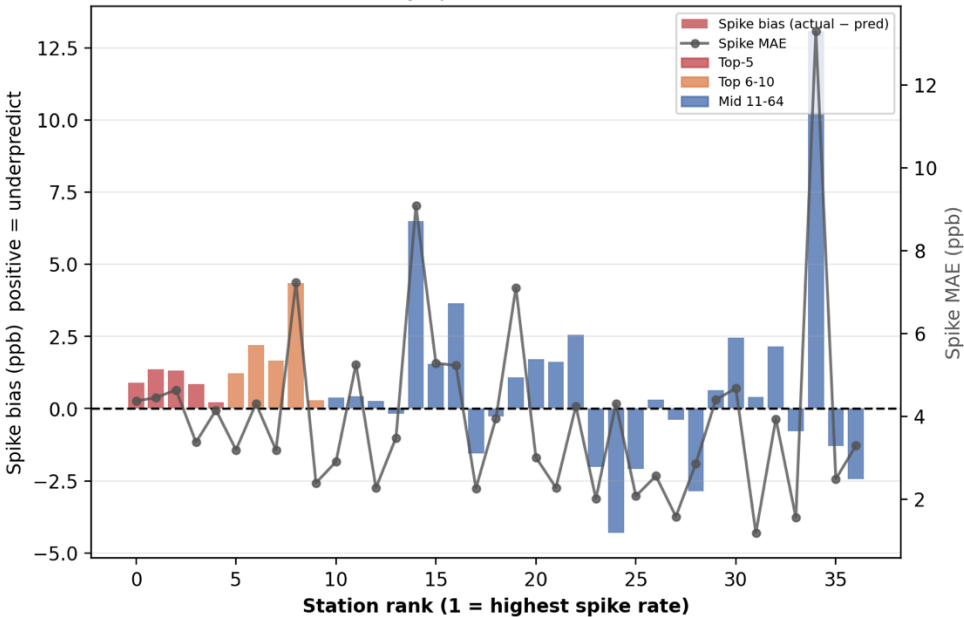


Shared Challenge: High-NO₂ Spikes Are Hard for Both Models

Our Model: Spike-Only Regressor (spike rows only)



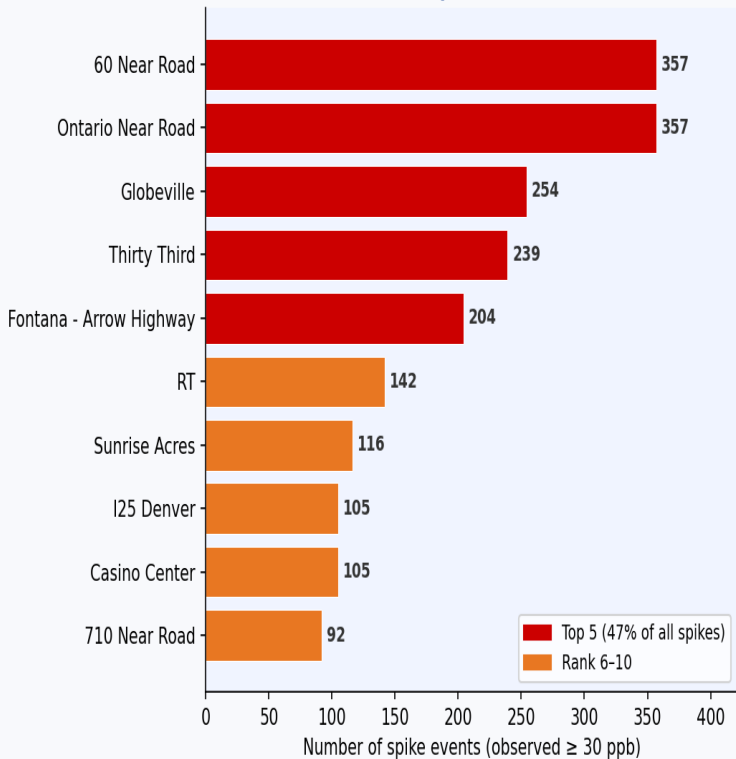
(4) Per-station spike bias and MAE (ordered by spike rate; colour = tier)



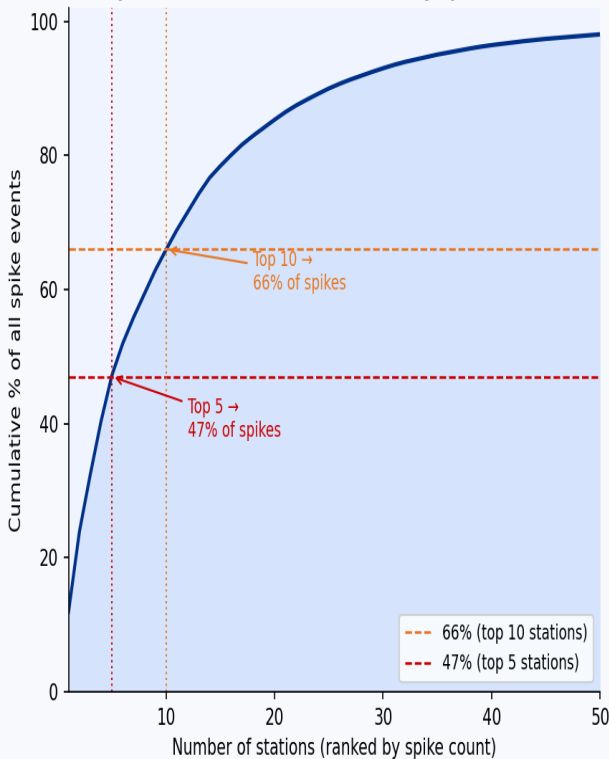
Graph Mamba: Spike Errors Also Cluster at a Few Stations



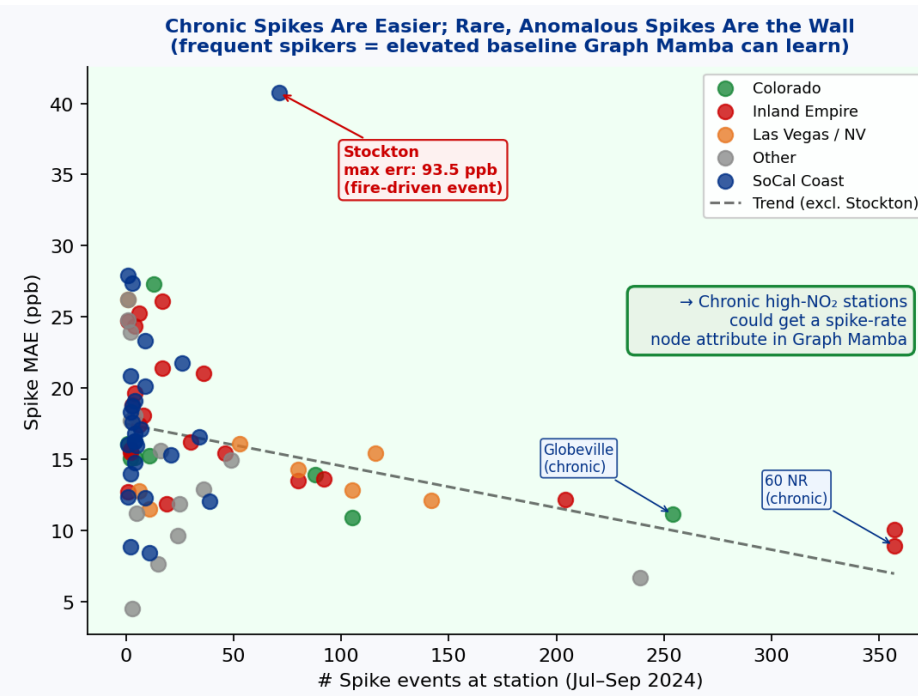
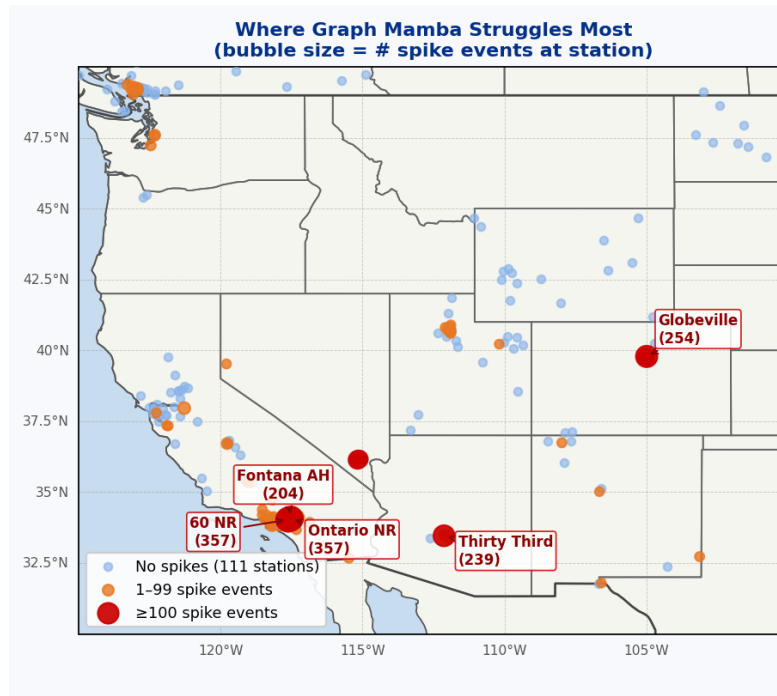
Top-10 Stations by Spike Event Count (Graph Mamba)



Spike Concentration: A Few Stations Drive Most Events (Graph Mamba – 77 stations have any spike, 188 total)



What I Found in Your Work



Insight #1: Same spike stations, different model

Graph Mamba's top-5 error stations (60 Near Road, Ontario NR, Globeville) are the same near-road, heavy-emission sites driving our spike model. This is a signal neither model has fully solved yet.

→ Station spike-rate priors could be added as graph node attributes in Graph Mamba

Insight #2: Both use TEMPO L3 V03, same test period

Graph Mamba predicts the continuous NO₂ field across all stations using a spatial graph + 8-scan Mamba sequence. Our model predicts whether and how big a spike will be at each station using per-station features (land use, emissions, met, climatology)

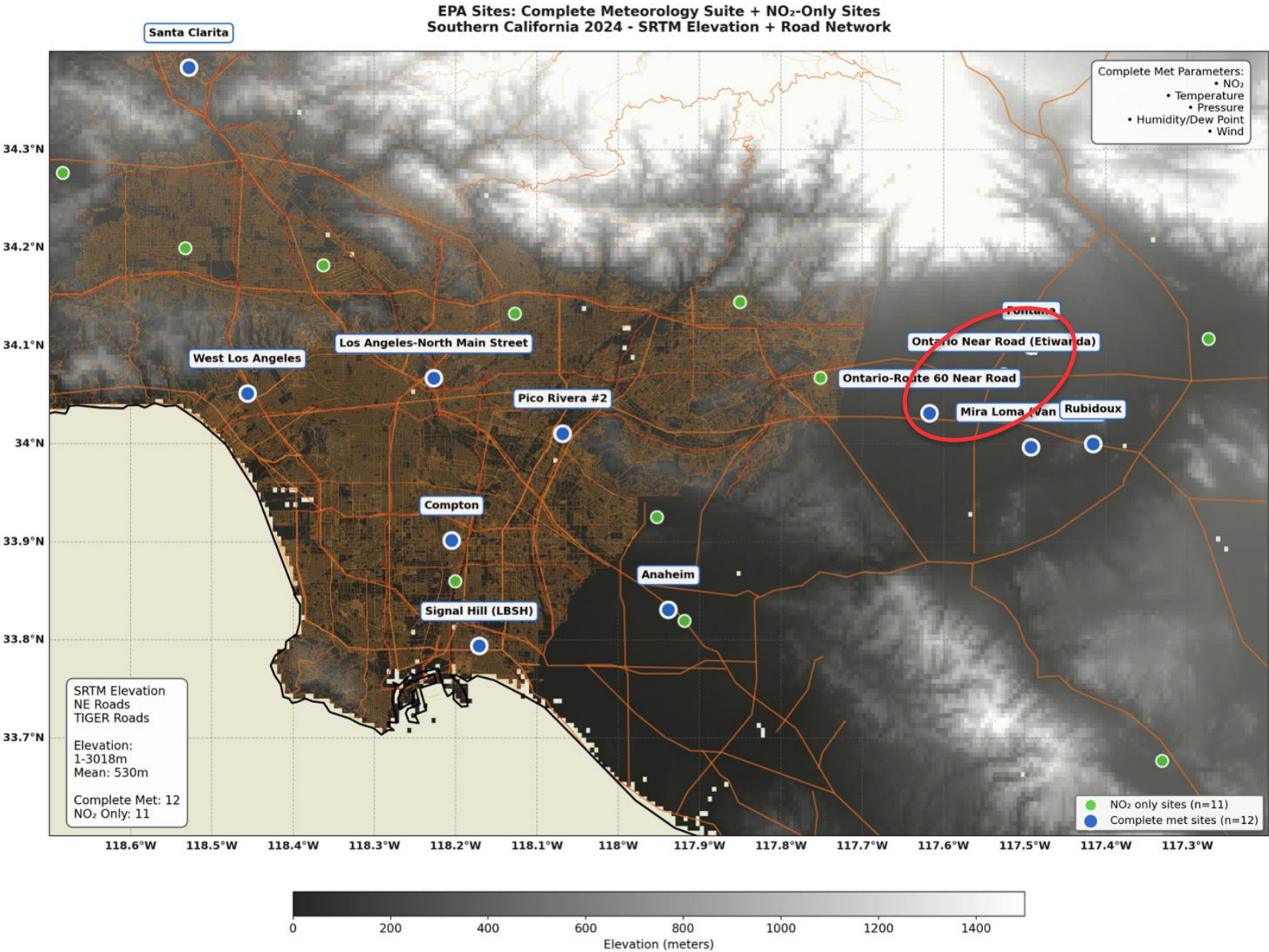
→ I might step Graph Mamba's interpolated field determine if it can help my model

Insight #3: Spike-aware fine-tuning opportunity

Graph Mamba has no spike-specific loss term. Its bias on ≥30 ppb rows is -12.1 ppb (systematic under-prediction). A two-stage approach — our hurdle classifier gating a spike-aware Graph Mamba branch — could directly fix this.

→ Try adding a spike detection head on top of Graph Mamba's node embeddings

LA Basin: Complete Meteorology + NO₂-Only Sites with SRTM Elevation



Key Takeaways: XGBoost Spike Detection vs. Graph Mamba

Our XGBoost Pipeline

✓ Strong spike detection

79% Top-1% Purity with lag NO₂ + emissions features

✓ Magnitude trend

Bias reduced from +10.4 → +3.8 ppb on spike rows

✓ Interpretable features

Can attribute predictions to emissions, met, climatology

✗ Magnitude limit

R² on spike rows still low; top-5 stations drive most error

✗ Single season / region

Trained & tested on California 2021 only

Graph Mamba Interpolation

✓ Broad spatial coverage

191 stations across North America, R² = 0.594 overall

✓ Low bias overall

+0.171 ppb mean bias across all concentration levels

✓ Graph-based structure

Captures spatial correlations between nearby stations

✗ High-NO₂ regime

MAE jumps to 13 ppb for ≥ 30 ppb events — same wall

✗ Test metrics NaN issue

val_fraction = 0 in config; need fix for auto reporting